

Q.1)

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a) The paper provides a chronological history of ERP, from 1960's ROP to 1970's MRP and MRP II in 1980's.

b) By creating "niche products and vendors" for specific industries. Examples of these specific segments include refineries, hospitals, automotive assembly and law offices. Preconfigured software modules would incorporate best practices and standard business processes.

c) Yes, through use of data mining and intelligence tools (expert systems that can suggest business decisions), advanced planning system (utilization of optimization), Simulation, push information (systems will offer information based on user-defined interests rather than user search).

Q2)

a) FDD:

Sequence $\rightarrow B \rightarrow E \rightarrow C \rightarrow D \rightarrow A$

Job	Proc. time	Due Date	Completion	Lateness	Tardiness
B	3	4	3	-1	0
E	4	6	7	1	1
C	1	7	8	1	1
D	6	10	14	4	4
A	2	12	16	4	4

total flow time

maximum tardiness
total tardiness

b) $D \rightarrow A \Rightarrow$ cause 8 tardiness
 $\Rightarrow$ Switch

Job A finishes at $8(6) + 2 = 10$

Tardiness for A become 0.

Job D finishes at $10 + 6 = 16$

Tardiness for D become 6

Before switch - After Switch = 2
(8) (6)

Total tardiness = $10 - 2 = 8$

$$c) w_j, v_j = 1 \quad k=2$$

$$\text{Formula: } I_j(t) = \frac{1}{p_j} \exp\left(-\frac{\max(d_j - p_j - t, 0)}{k \cdot \bar{p}(t)}\right)$$

$$\bar{p}(0) = (2+3+1+6+4)/5 = 3.2$$

$$I_A(0) = \frac{1}{2} e^{(-10/6.4)} = 0.105$$

$$I_B(0) = \frac{1}{3} e^{(-1/6.4)} = 0.295$$

$$I_C(0) = \frac{1}{6} e^{(-6/6.4)} = 0.391 \checkmark$$

$$I_D(0) = \frac{1}{6} e^{(-4/6.4)} = 0.089$$

$$I_E(0) = \frac{1}{4} e^{(-2/6.4)} = 0.183$$

$$C \text{ is selected} \Rightarrow \bar{p}(1) = (2+3+6+4)/4 = 3.75$$

$$I_A(1) = \frac{1}{2} e^{(-9/7.5)} = 0.151$$

$$I_B(1) = \frac{1}{3} e^{(0/7.5)} = 0.333 \checkmark$$

$$I_D(1) = \frac{1}{6} e^{(-3/7.5)} = 0.112$$

$$I_E(1) = \frac{1}{4} e^{(-1/7.5)} = 0.219$$

$$B \text{ is selected} \Rightarrow \bar{p}(4) = (2+6+4)/3 = 4$$

$$I_A(4) = \frac{1}{2} e^{-4/8} = 0.236$$

$$I_D(4) = \frac{1}{6} e^0 = 0.167$$

$$I_E(4) = \frac{1}{4} e^0 = 0.25 \checkmark$$

$$E \text{ is selected} \Rightarrow \bar{p}(8) = 2+6/2 = 4$$

$$I_A(8) = \frac{1}{2} e^{(-2+8)} = 0.389 \checkmark$$

$$I_D(8) = \frac{1}{6} e^0 = 0.167$$

$$A \text{ is selected} \Rightarrow \text{new time} \quad 8+2=10$$

Schedule $\Rightarrow C - B - E - A - D$

Total for dress

$$E \Rightarrow 2$$

$$D \Rightarrow 6$$

$$\text{others } 0$$

$$\Rightarrow 2+6=8$$